

Multilevel Determinants of Player Market Value in Elite European Soccer

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Abstract

The valuation of elite soccer players has evolved into a global financial industry exceeding €7 billion annually, yet traditional models often fail to capture the dynamic movement of players between clubs. Using a cross-classified multilevel model on 7,023 observations (2013–2024), we partition market value variance into individual reputation and club context. We identify a distinct market cycle and a strict career arc peaking at age 27. Notably, club status drives valuation more than previously understood, with key starters commanding a 42% premium over rotation players. We also identify a diversity premium, where clubs with higher international representation generate higher player valuations, contrasting with a systematic 17% discount for domestic talent.

Keywords: soccer economics, multilevel modeling, player valuation, club prestige, UEFA coefficient, sports analytics

1 Introduction and Research Motivation

The valuation of elite athletes in the European soccer transfer market has evolved from a subjective scouting exercise into a global financial industry exceeding €7 billion annually. In this high-stakes environment, accurate valuation is critical for club solvency and competitive balance. In this study, we operationalize market value using crowd-sourced estimates from Transfermarkt. This metric has been widely validated in sports economics literature as a robust proxy for actual transfer fees and serves as a leading indicator of player market valuation based on demand.

Research in sports economics has long sought to isolate the determinants of transfer value. While Frick (2007) established that performance metrics like goals and assists are primary drivers, valuation follows a strict “Inverted-U” biological trajectory, typically peaking between ages 27 and 28. Beyond on-pitch output, Franck and Nüesch (2012) demonstrated a “Superstar Effect (Rosen, 1981),” where brand visibility magnifies value at the top of the wage distribution. However, existing models frequently treat players as static units nested within a single club. This approach ignores the reality of modern soccer in which players are mobile assets who accumulate value from both their portable human capital (i.e., experience, skills, reputation) and the institutional prestige of their current club. Neglecting this cross-classified structure—where player and club effects intersect rather than nest—misestimates the true drivers of market value. When we apply a cross-classified multilevel model to a decade of transfer data, we disentangle individual reputation from club context to address four critical questions:

1. **Market valuation evolution:** Distinguishing real value growth from inflation.

2. **Career arc:** Identifying the precise biological window where physical peak and experience intersect.
3. **Club status:** Testing whether relative playing time signals value more reliably than raw counting stats.
4. **Contextual premiums:** Quantifying the valuation impact of club diversity and domestic status in a globalized labor market.

2 Method

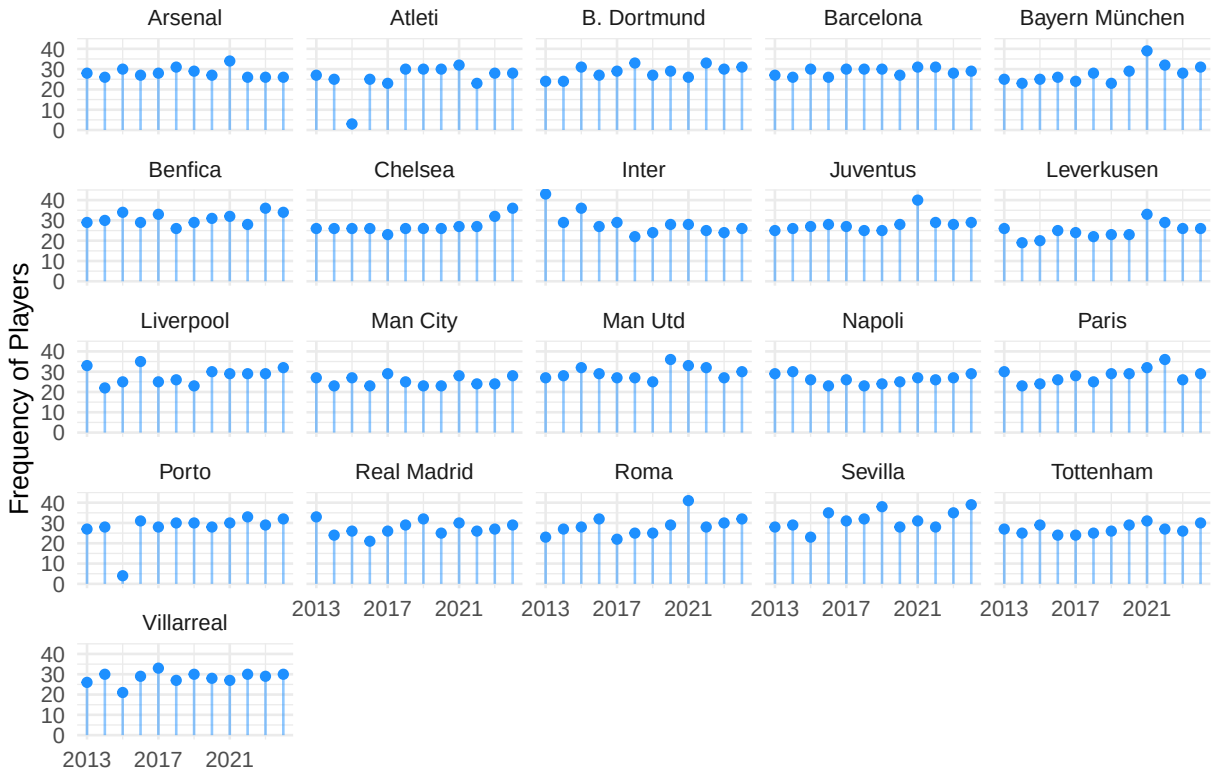
Data Source. We obtained data from [Transfermarkt](#) and [Union of European soccer Association](#) (UEFA) official rankings, which are both publicly available websites. At the time of this study, Transfermarkt does not have free API access to their data repository. Instead, the R package [worldsoccerR](#) allows us to scrape data from their website. However, there are several limitations to this package such as timeout requests and failure to capture dynamic JavaScript tables on webpages. We found the necessary data through Kaggle [soccer Data from Transfermarkt](#), which allowed us to focus on the modeling aspect of this project following data restructuring and predictor variable engineering.

Sample Characteristics. We used the 10-year averaged UEFA coefficient rankings¹ to select the top 21 European soccer teams as our first stage sampling. In our second stage nonrandom sampling, we selected soccer players that played for a full season in one of the 21 clubs from 2013 to 2024. This resulted in 2,215 male players representing 7,168 seasons.

¹UEFA coefficient rankings are a points-based system used by European soccer’s governing body to rank national associations and clubs based on their performance in European competitions, which determines qualification spots and seeding for tournaments like the Champions League and Europa League.

However, after examining missing data (which mostly came from missing information on citizenship status), we were left with 2,159 players representing 7,023 seasons. This represents a 2.5% loss of player data and a 2% loss in season data. The majority of players across clubs and years were in their mid-20s ($M = 26.5$, $SD = 4.6$). In this sample, players on average played 3.2 ($SD = 2.9$) seasons. If we were to drop players that had less than two seasons to examine player trajectories, we would lose 62% of our sample. Figure 1 shows the number of players in each of the 21 clubs across the years.

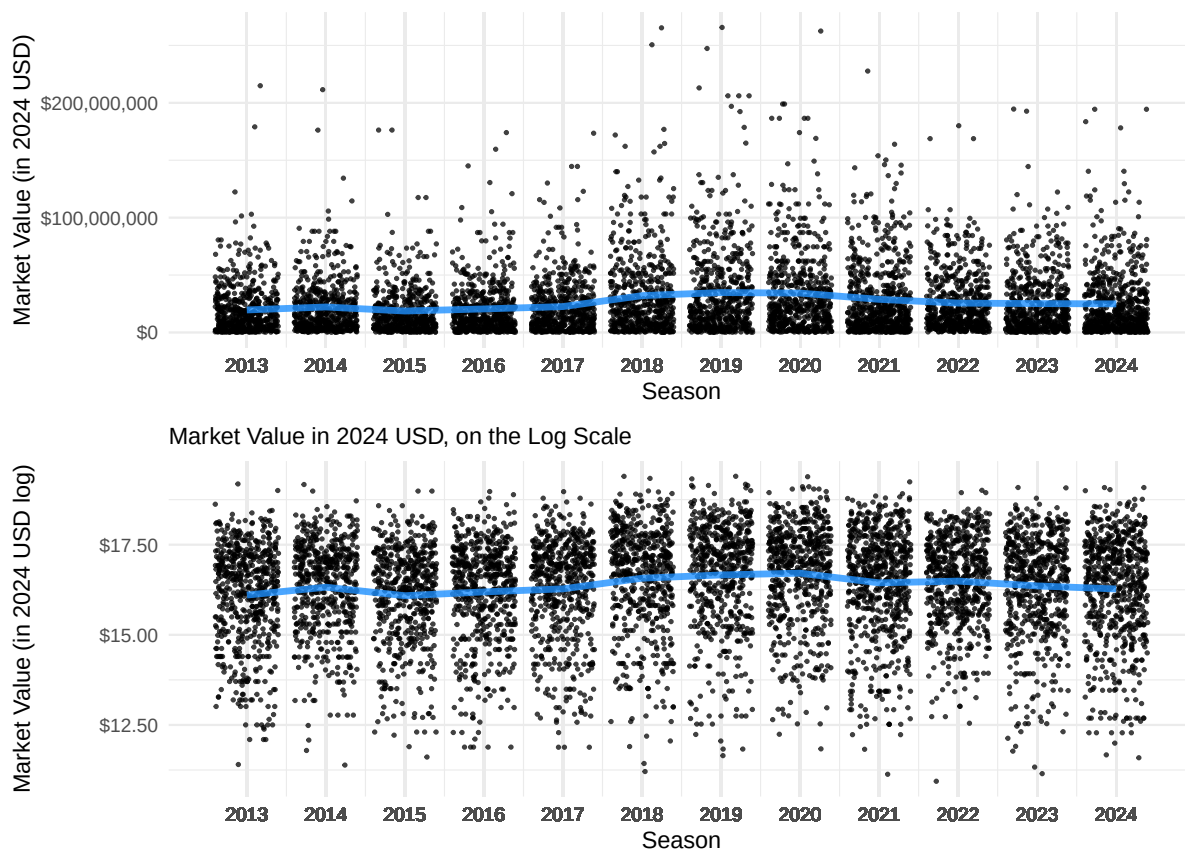
Figure 1. On Average Each Team Contains about 30 Players Per Season



Variable Selection and Normalization. Our outcome of interest is a player's yearly market valuation, which we convert from euros to US dollars. Market valuation in euros varies by month, so we first calculated the yearly average for each player prior to transforming to USD. We log-transform market value to 2024 USD because financial data follows a Pareto

distribution rather than a normal distribution. Keeping the outcome on the raw scale would result in extremely large residuals for high-value players and small residuals for low-value players, which violates the assumption of homoscedasticity. Log-transforming the outcome compresses the extreme outliers by pulling the long right tail to make the distribution more normal. Therefore, this adjustment allows us to compare the real economic power of a player in 2013 versus 2024 without the noise of currency fluctuation.² Figure 2 shows that market value has substantial variability across years, yet most values are at the lower end, which is common for salary data. When we log-transform our outcome, we bring the lower values up and the higher values down.

Figure 2. Market Value in 2024 USD, Blue line: Average Season Market Value



²We handle exchange rate fluctuation by using historical exchange rates to convert our values to USD. We also adjust market value to 2024 USD because 10 million dollars in 2013 is not the same as 10 million in 2024 due to inflation. By keeping the raw values, we would lose information about real economic value.

A list of variables we used for modeling players’ market values are shown in Appendix 1 alongside our justification for how and why we centered each fixed effect, such as ensuring meaningful model intercepts and solving convergence (e.g., degenerate Hessian). We applied the following centering and normalization/transformation strategies:³ We used the natural log of inflation-adjusted market value (`usd_revenue_2024_log`). We centered season year at the minimum value, making 2013 equal to 0 (`season_year_c`). We applied grand-mean centering to age (`age_c`) and height (`height_c`). We standardized minutes played (Z-score) relative to each specific club-year. This isolates a player’s status as a starter or bench player relative to their current teammates and removes the noise of team level rotation policies. We also grand-mean centered the proportion of foreign players (`foreign_players`). We included region, position, and citizenship status (coded as 1 for citizens of the club country). We centered the number of goals and assist by the four positions of players (e.g., defender, midfielder, goalkeeper, attacker). This ensured that a goal from a defender is weighted differently than a goal from an attacker since they are more likely to strike a goal (i.e., goalies are classified as defenders in these data).

3 Modeling Approach

We employ a cross-classified multilevel linear model to partition variance across observations (Level 1), players (Level 2a), and clubs (Level 2b). This structure explicitly accounts for players transferring between clubs and thus treating player and club as cross random factors

³*Data Assumptions:* We assume that variables such as height and positions are time invariant for the duration of this study. At the club level, we assume mutually exclusiveness in that players can only belong to one club in a given year. Although a player could have switched clubs mid-season, we keep this constant for model simplicity. We also assume that the relationship between age and log-market value is non-linear and parabolic so we introduce a quadratic term on age. Because we excluded observations with missing data (listwise deletion), we assume the data are Missing Completely at Random (MCAR) or that the proportion of missing cases (~2%) is negligible and does not introduce systematic bias.

rather than a strict three level hierarchy. We found that in this sample, almost 20% of players had been in two to seven (17% in two clubs) clubs which was the basis for fitting a cross-classified model to these data. Figure 3 is a visual of a cross-classified model example of three players that either move through clubs or remain in the same club throughout their career. Players can change teams over their span of their careers meaning that the player context can change. They may be a star player in one team and a bench player on a different team that they switch to. In a cross-classified model, the player is the constant unit (i), but the club (j) changes over time (t). If player (I) plays for club (j1) in year (t0) but switches to club (j2) in year (t1) the fixed effect in the model, such as minutes played, updates the players status, as opposed to having static or average predictors. Club attributes, such as foreign players and UEFA scores, ensures that when player move clubs, the model swaps the values of X_{j1} , for X_{j2} . Player's characteristics, for instance age and height, move with the player across time.

Figure 3. Cross-Classified Data Structure: Players move between clubs over time



In the overall spaghetti plot in Figure 4.a, grouped by club, they consistently show differences and are not constant across years. This means between-club differences in market values exist, and their trajectory change over time as well, indicating slopes might vary across clubs. We also plot in Figure 4.b and 4.a. what appears to be a quadratic and cubic

relationship between market value and time. This provides us with enough justification to include test these terms in our models.

Figure 4.a. Market Value Club Trajectories

Blue line: Average time point Market Value

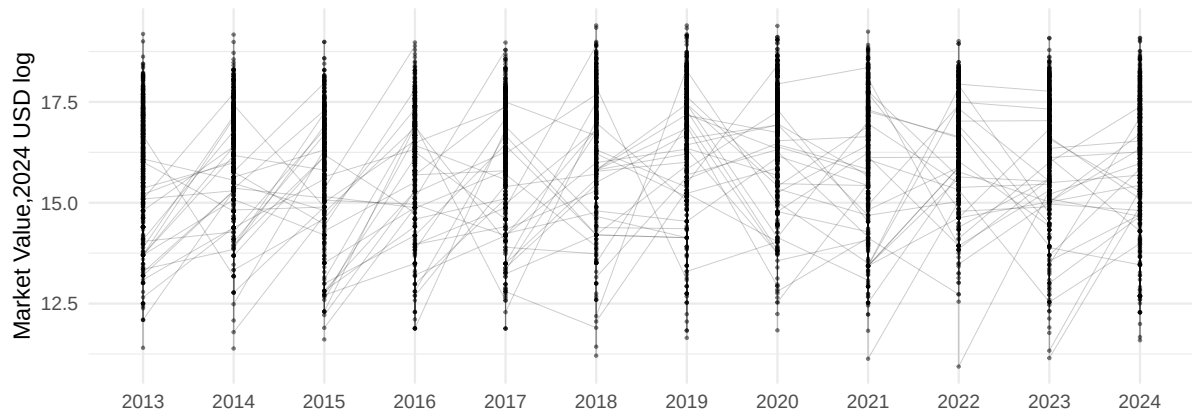


Figure 4.b. Club Market Value Trajectories, Quadratic Relationship

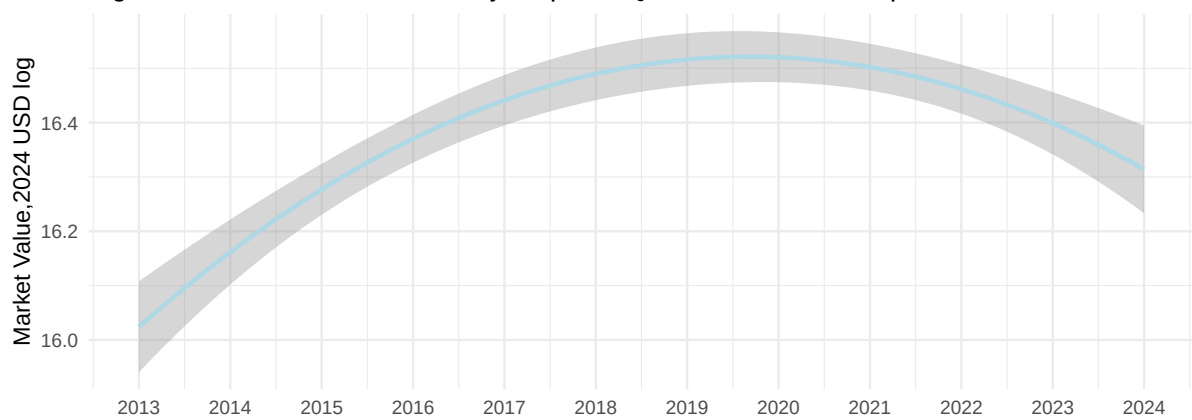
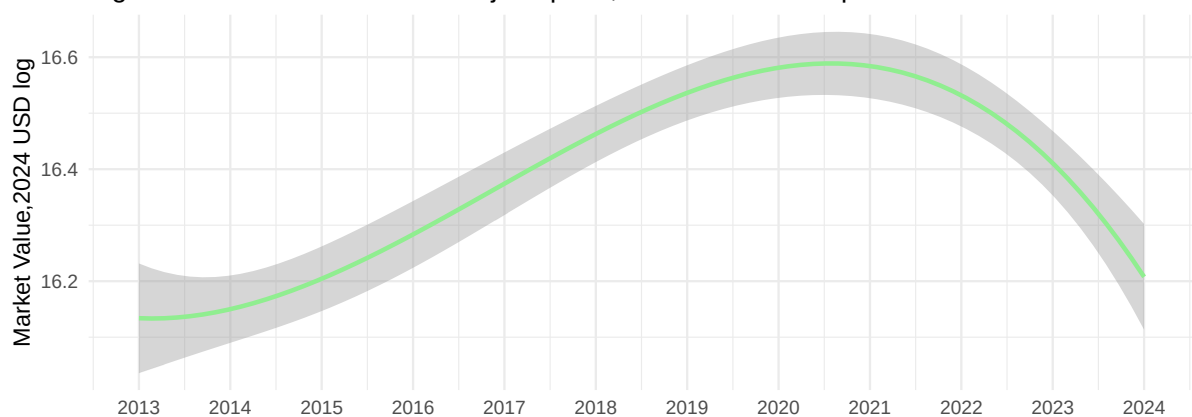


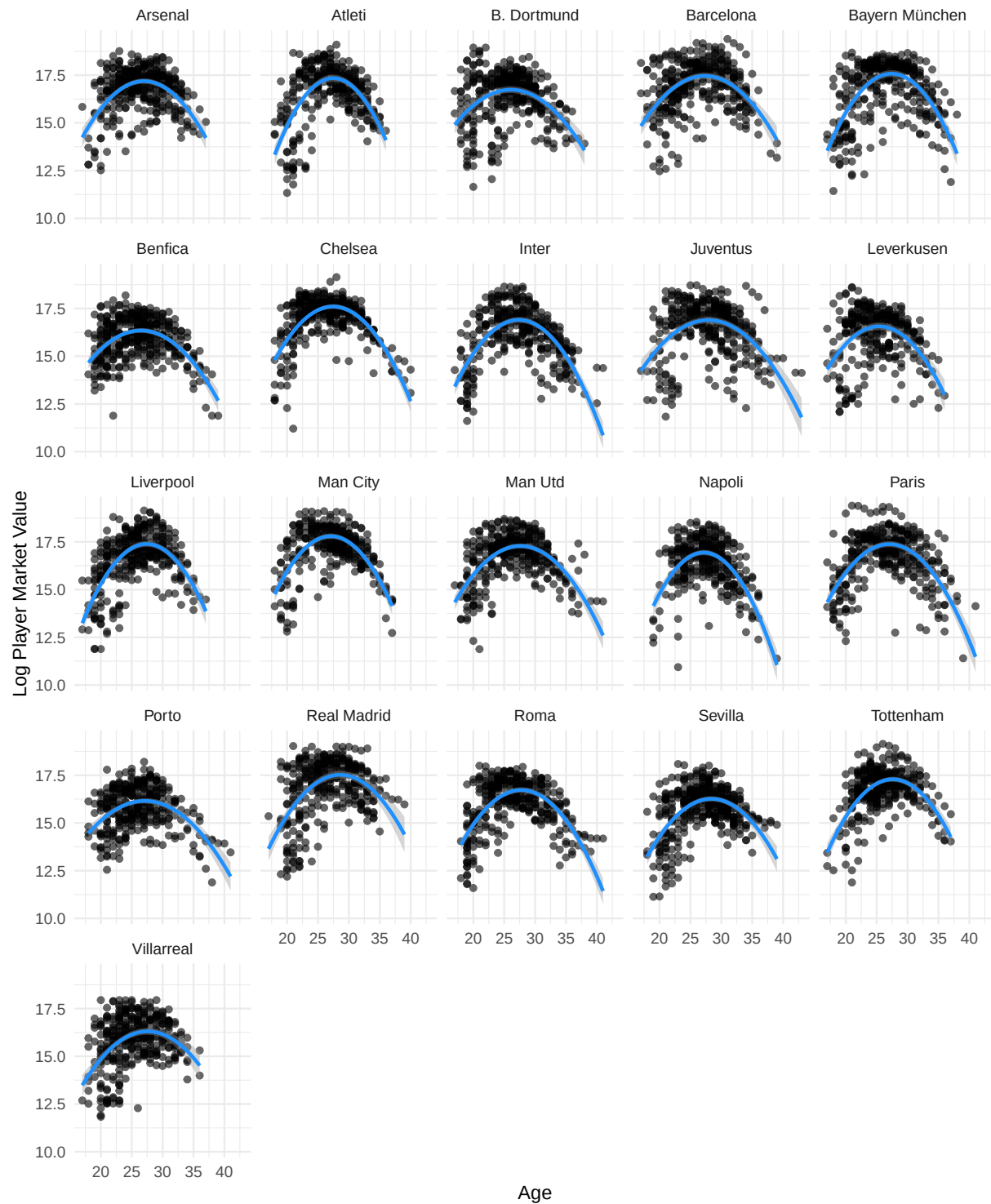
Figure 4.b. Club Market Value Trajectories, Cubic Relationship



We plot age by yearly market value in 2024 USD (on the log scale) in Figure 5. for each

team and observe almost an upside down u shape. As players get older you are valued less in the market, same for younger players. Again, this indicates to us that testing a quadratic term on age should be included in our modeling plan.

Figure 5: Age and Market Value (log) follow a Quadratic Relationship



3.1 Model Specification

To formally describe our analysis, we utilize a hierarchical notation system that decomposes the model into observation, player, and club levels. This analysis clarifies how the cross-classified structure (i players crossed with j clubs) contributes to the variance in market value. Here we express our final model.

Level 1: Observation Model (Time-Varying Effects)

At the observation level (time t , player i , club j), the log-transformed market value is a function of a player-club specific intercept (π_{0ij}), temporal trends, biological aging, and dynamic performance metrics:

$$\begin{aligned} \log(\text{Revenue})_{tij} = & \pi_{0ij} \\ & + \pi_{1j}(\text{Year}_t) + \pi_2(\text{Year}_t^2) + \pi_3(\text{Year}_t^3) \\ & + \pi_4(\text{Age}_{it}) + \pi_5(\text{Age}_{it})^2 \\ & + \pi_6(\text{Mins}_{Z,it}) + \pi_7(\text{Goals}_{it}^*) + \pi_8(\text{Assists}_{it}^*) \\ & + \pi_9(\text{Red}_{it}) + \pi_{10}(\text{Yellow}_{it}) \\ & + \epsilon_{tij} \end{aligned}$$

where $\epsilon_{tij} \sim N(0, \sigma_\epsilon^2)$.

Level 2a: Player Model (Time-Invariant Traits)

The player-club intercept (π_{0ij}) is determined by static physical and demographic characteristics, citizenship status relative to the club, and a player-specific random effect:

$$\pi_{0ij} = \beta_{0j} + \gamma_1(\text{Height}_i) + \gamma_2(\text{Citizen}_{ij}) + \sum_p \gamma_p \mathbb{1}_{\text{Position}_i=p} + \sum_r \gamma_r \mathbb{1}_{\text{Region}_i=r} + u_{0i}$$

where $u_{0i} \sim N(0, \sigma_u^2)$.

Level 2b: Club Model (Contextual Effects)

The club-level baseline (β_{0j}) and the linear time slope (π_{1j}) are modeled as functions of club internationalization and club-specific random effects:

$$\beta_{0j} = \delta_{00} + \delta_{01}(\text{Foreign}_{tj}) + v_{0j}$$

$$\pi_{1j} = \delta_{10} + v_{1j}$$

$$\pi_2 = \delta_{20} \quad (\text{fixed across clubs})$$

$$\pi_3 = \delta_{30} \quad (\text{fixed across clubs})$$

where $\begin{pmatrix} v_{0j} \\ v_{1j} \end{pmatrix} \sim N(\mathbf{0}, \Sigma_{club})$, with $\Sigma_{club} = \begin{pmatrix} \sigma_{v_0}^2 & \sigma_{v_0 v_1} \\ \sigma_{v_0 v_1} & \sigma_{v_1}^2 \end{pmatrix}$.

Interpretation of Cross-Classification:

Player effects (u_{0i}): Capture persistent individual heterogeneity (e.g., reputation, marketability) that follows a player across clubs.

Club effects (v_{0j}, v_{1j}): Capture the baseline prestige and temporal trajectory of the club itself, which affects all players on the roster.

Cross-classified structure: A player's market value at time t depends on both their intrinsic

“brand” (u_{0i}) and the club they are currently with ($v_{0j} + v_{1j} \cdot \text{Year}_t$).

4 Results

We fitted a series of nested multilevel models to identify the optimal specification. We started with a null model containing only random intercepts for players and clubs to establish baseline variance partitioning. We then systematically added temporal trends (linear, quadratic, and cubic), random slopes for club trajectories, player-level predictors, and finally club-level contextual factors. Model selection and comparison was guided by Likelihood Ratio Tests (LRT) to evaluate the statistical significance of nested model improvements. To balance goodness-of-fit against model complexity, we also consulted the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Models yielding significant LRT results ($p < .05$) and lower values on both information criteria were retained for subsequent analysis. We describe each model in Appendix 2, Table 2 alongside ICC, AIC, and BIC values.

Appendix 3 contains all of our modeling results. Our random intercept only model reveals that `player_id` accounts for about 75% of market value while `club_id` accounts for 3%. We test whether a random slope on year is needed the LRT test⁴ reveals that the random slope provides a significantly better fit than the reduce model, the quadratic term is needed for this data. We also tested a random slope of year within club which did not converge, probably due to a large proportion of players only play 1-2 seasons in our data. Overall, our

⁴The alternative hypothesis, $H_A = \tau_{11} > 0$ since variance should be larger than 0. To test this hypothesis we use the likelihood ratio test (LRT) simply by $-2[\log \text{likelihood}_{reduced} - \log \text{likelihood}_{full}]$ to examine the difference. We use a mixture of χ^2 1 and 2 distributions to compare the full model with two random effects against a reduced model with one random effect.

final model included player characteristics, performance metrics, and club characteristics, and we select Model 4 because we found a non significant LRT test different between when we include club characteristics versus only player characteristics(Model_3.2) despite Model 4 having a slightly larger AIC and BIC than Model 3.2.

IN DRAFT

Our model reveals a significant cubic trend ($p < .001$) in market evolution. As shown in Figure 6A, the market followed an S-shaped trajectory. We observed stagnation from 2013 to 2014, followed by a rapid inflationary boom from 2015 to 2019, and then a post-2020 plateau. This confirms that soccer market inflation is cyclical rather than constant.

We observe a strong quadratic effect for age ($p < .001$). Figure 6B illustrates the inverted-U shape, where value peaks during a player's prime (approximately age 27) and declines significantly in the veteran stage.

Our results show that squad status is a massive driver of valuation. Because we standardized minutes played, the coefficient (approximately 0.35) indicates that a one standard deviation increase in playing time is associated with a 42% increase in market value. This confirms that being a starter is financially distinct from being a rotation player.

We find evidence of a diversity premium. A 10 percentage point increase in a club's foreign player proportion increases average player value by 2%. Conversely, we find a domestic discount where homegrown citizens are valued 17% lower than foreign imports.

LRT Statistic: 66.6861

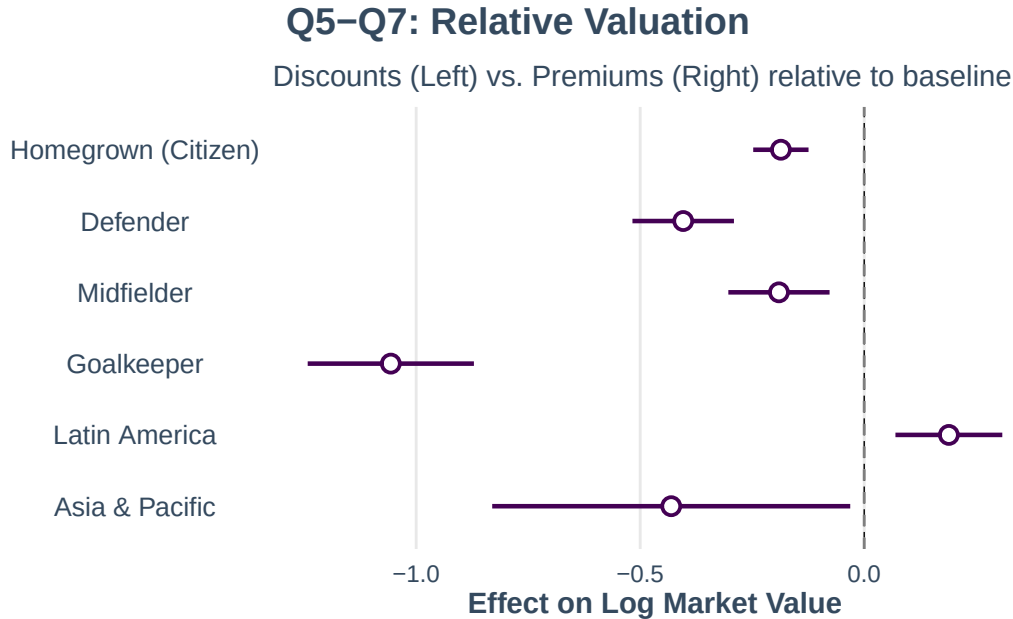
Parameter Diff: 2

Distribution: Mixture of ChiSq(1) and ChiSq(2)

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
26	14362.95	14541.23	-7155.476	14310.95	NA	NA	NA
28	14361.89	14553.88	-7152.944	14305.89	5.064713	2	0.07947152

P-Value: < 0.0001

Below are marginal effects for our categorical variables that we need to unpack.



5 Club-Level Heterogeneity

Figure 7 displays the estimated club random effects from our cross-classified model. We find substantial heterogeneity in both baseline valuations and temporal trajectories.

5.1 Baseline Premiums and Discounts

Manchester United exhibits the largest effect. When controlling for age, position, and performance, United players are valued 68% higher than league average. The 95% confidence interval ranges from 31% to 76%. Barcelona shows a 56% premium (95% CI from 29% to

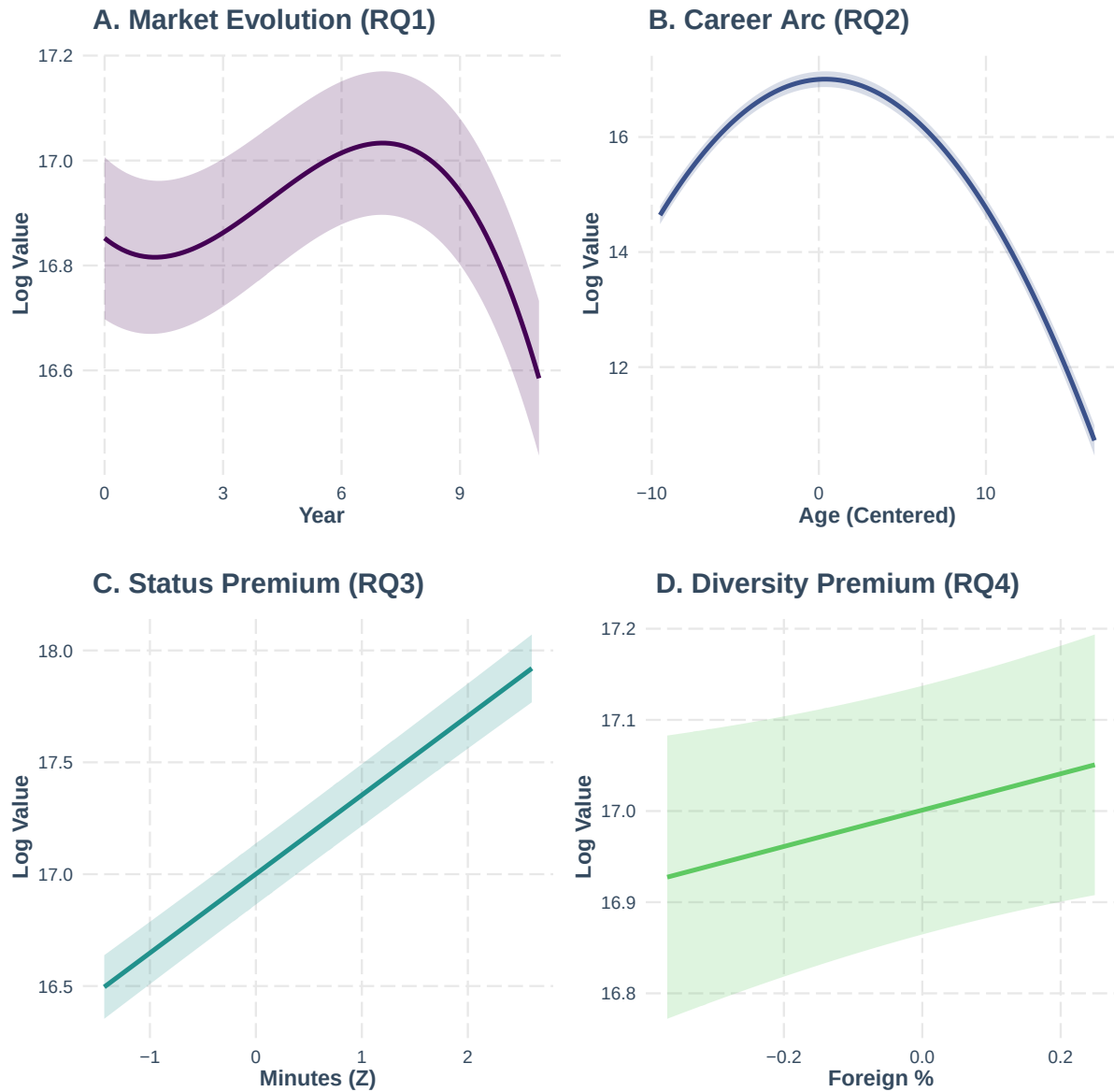


Figure 1: Key drivers of market value. Panel A shows the cubic market trend over time. Panel B displays the quadratic career arc with peak around age 27. Panel C illustrates the squad status premium for starters. Panel D demonstrates the diversity premium for international rosters.

93%), while Paris Saint-Germain shows a 26% premium (95% CI from 6% to 53%). Chelsea, Manchester City, and Bayern Munich also show significant premiums of approximately 22% to 25%.

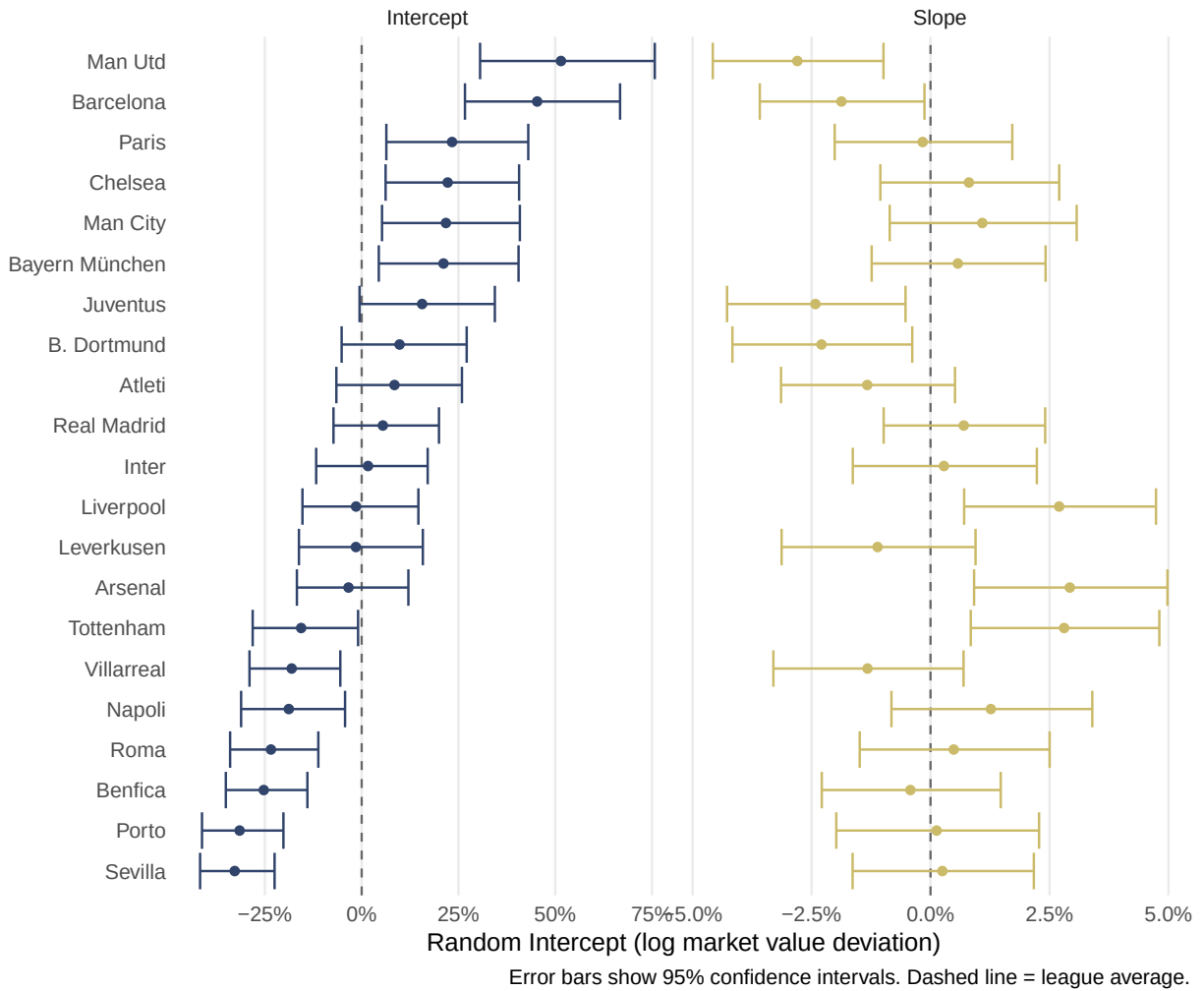
Conversely, seven clubs show significant discounts. Sevilla faces the steepest penalty, with players valued 28% below league average (95% CI from -42% to -23%). Porto shows a 27% discount, Benfica shows 22%, Roma shows 21%, Napoli shows 17%, Villarreal shows 16%, and Tottenham shows 14%.

5.2 Diverging Temporal Trajectories

Our random slopes reveal that club fortunes diverged over the study period. Three clubs showed significantly faster appreciation than league average. Arsenal showed growth of 2.9% per year (95% CI from 0.8% to 5.0%), Liverpool showed 2.7% per year (95% CI from 0.7% to 4.8%), and Tottenham showed 2.8% per year (95% CI from 0.8% to 4.8%). These trajectories coincide with improved sporting performance and, for Liverpool, a Champions League victory in 2019.

Four clubs experienced significantly slower growth or faster decline. Manchester United declined 2.8% per year (95% CI from -4.5% to -1.0%), Borussia Dortmund declined 2.3% per year (95% CI from -4.1% to -0.4%), Juventus declined 2.3% per year (95% CI from -4.1% to -0.5%), and Barcelona declined 1.8% per year (95% CI from -3.5% to -0.1%). Despite high baseline premiums, these clubs saw their market advantages erode over time.

Figure 7.Club–Level Market Value Premium



6 Findings

Our study demonstrates that player market value is driven primarily by individual reputation and squad status, but is significantly moderated by club context and market cycles. Our cross-classified approach reveals that 78% of variance can be attributed to persistent player effects while accounting for dynamic club contexts.

While our cross-classified model demonstrates strong predictive power (R-squared approximately 0.89), several limitations warrant consideration. First, although the likelihood ratio test supported including random slopes for club trajectories, we should interpret the variance estimates with caution given the relatively small number of unique clubs ($N = 21$). Second, our diagnostic plots indicated deviations from normality in the extreme tails of the residual distribution. While our large sample size ($N = 7,023$) ensures robust estimation of fixed effects, future iterations could employ robust standard errors or a Gamma GLMM to better address these extremes. Finally, predictors such as minutes played are likely endogenous, since better players naturally accrue more playing time. We should interpret our estimates as associations reflecting squad status rather than purely causal effects.

Ultimately, our research highlights the necessity of moving beyond static, single-level analyses to capture the true complexity of the soccer transfer market. By rigorously modeling the cross-classified intersection of individual player trajectories and dynamic club contexts, we provide a more accurate lens for valuing talent in a hyper-inflated economy. As the industry continues to professionalize, such multidimensional frameworks (which account for the interplay of human capital, institutional prestige, and market cycles) will remain indispensable for clubs seeking to distinguish genuine value from market noise.

7 References

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8 Appendix 1. Modeling Predictors and Centering Strategies

Table 1: Table 1. Data Structure for fixed effects, except year as it is a random slope

		Centering	
		Strategy/ Form	
Variable Type	Variable	Factor	Justification
Time	Season Year	Min-Centering (+	Establishes a meaningful
		Quadratic)	starting intercept (2013 =0)
Contextual, Club	UEFA Coefficient	Year-Mean	Isolates relative prestige;
		Centering	removes historical inflation
	Foreign Players Proportion	Grand-Mean	Moves baseline to “Average Club” (since 0% is unlikely)
Player Characteristics	Age	Grand-Mean (+	Captures absolute career
		Quadratic)	stages (Prospect → Prime → Veteran)
	Height (in cm)	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average

		Centering	
		Strategy/ Form	
Variable Type	Variable	Factor	Justification
Player Demographic	Citizen of Club	None (Raw	Keeps “Non-Citizen” as
	Country	binary 0/1)	clear reference group. Separates individual status from club composition (foreign_players)
Player Performance	Region of origin	Reference group, Europe	Control variable for where player is from
	Goals	Position- Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position- Centering	Preserves natural 0 and marginal value of 1 assist interpretation
	Red Card	raw season count	Meaningful 0 interpretation
	Yellow Card	raw season count	Meaningful 0 interpretation
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player

9 Appendix 2. Modeling Plan

Models	Specification / Additions	Purpose & Research Justification	ICCplayer/ club	AIC	BIC
Model 1	Unconditional Means Model (Random Intercepts only)	Calculate ICC and partition variance between players and clubs before adding predictors	.746/.034	20105.1893	20132.6171
Model 2.1	+ Fixed Effect: season_year_c	Test for a general linear trend in market value growth	.756/.033	20052.802	20087.087
Model 2.2	+ Fixed Effect: season_year_c_sq (Year ²)	Test for non-linear market dynamics (e.g., acceleration or decelera- tion/plateaus)	.757/.033	20046.978	20088.119

Model 2.3	+ Fixed Effect:	Test for complex	.758/.033	20037.404	20085.402
	$\text{season_year_c_club}$	“S-shaped”			
	($Year^3$)	market			
		trajectories (e.g.,			
		booms)			
Model 2.4	+ Random	Test if market	.749/.051	19974.718	20036.430
	Slope:	value growth			
	season_year_c	rates vary			
	(at Club	significantly			
	Level)	between clubs			
Model 2.5	+ Random	Test if the growth			No
	Slope:	curvature varies			Convergence
	season_year_c_by	club			

Model 3.1	+ Player	Estimate the	.636/.096	17469.242	17640.666
	Fixed Effects:	marginal value of			
	Performance	performance			
	(Goals, Assists,	metrics and			
	etc.), Status	player			
	(Minutes),	characteristics			
	Demographics	controlling for			
	(Region,	time			
	Citizen),				
	Traits (Height,				
	Linear Age)				
Model 3.2	+ Fixed	Test the the	.775/.047	14507.097	14685.378
	Effect:	inverted-U career			
	age_c_sq	arc (Prospect →			
	(Age ²)	Prime →			
		Veteran)			
Model 4	+ Club Fixed	Test for	.773/.050	14520.484	14712.479
	Effects:	contextual club			
	foreign_players	effects			
	uefa_coefficients				

Final	Diagnostics &	Compare models
Model	Comparison	using LRT,
Selection		AIC/BIC, and
		residual
		diagnostics (QQ
		plots) to select
		the most
		parsimonious and
		robust fit

10 Appendix 3. R code for models and outputs:

10.1 We print model output here for review.

Printing fit results for Model 1

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20105.189
BIC	20132.617
Pseudo-R ² (fixed effects)	0.000
Pseudo-R ² (total)	0.780

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.821	0.069	228.895	26.989	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.321
club_id	(Intercept)	0.284
Residual		0.718

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.746
club_id	21	0.034

Printing fit results for Model 2.1

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.802
BIC	20087.087
Pseudo-R ² (fixed effects)	0.005
Pseudo-R ² (total)	0.791

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.991	0.072	221.312	32.906	0.000
season_year_c	-0.031	0.004	-8.113	6524.033	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.344
club_id	(Intercept)	0.282
Residual		0.709

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.756
club_id	21	0.033

Printing fit results for Model 2.2

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20046.978
BIC	20088.119
Pseudo-R ² (fixed effects)	0.006
Pseudo-R ² (total)	0.791

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.916	0.074	214.884	36.552	0.000
poly(season_year_c, 2, raw = TRUE)1	0.019	0.012	1.584	5756.860	0.113
poly(season_year_c, 2, raw = TRUE)2	-0.004	0.001	-4.451	5650.505	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.343
club_id	(Intercept)	0.282
Residual		0.708

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.757
club_id	21	0.033

Printing fit results for Model 2.3

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20037.404
BIC	20085.402
Pseudo-R ² (fixed effects)	0.007
Pseudo-R ² (total)	0.793

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.993	0.076	211.563	39.594	0.000
poly(season_year_c, 3, raw = TRUE)1	-0.093	0.025	-3.733	5460.096	0.000
poly(season_year_c, 3, raw = TRUE)2	0.022	0.005	4.153	5343.577	0.000
poly(season_year_c, 3, raw = TRUE)3	-0.002	0.000	-5.095	5343.972	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.346
club_id	(Intercept)	0.282
Residual		0.705

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.758
club_id	21	0.033

Printing fit results for Model 2.4

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19974.718
BIC	20036.430

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.982	0.089	180.211	28.024	0.000
poly(season_year_c, 3, raw = TRUE)1	-0.092	0.026	-3.533	1018.972	0.000
poly(season_year_c, 3, raw = TRUE)2	0.023	0.005	4.304	5311.323	0.000
poly(season_year_c, 3, raw = TRUE)3	-0.002	0.000	-5.290	5311.733	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.351
club_id	(Intercept)	0.353
club_id	season_year_c	0.039
Residual		0.697

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.749
club_id	21	0.051

Printing fit results for Model 3.1

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17469.242
BIC	17640.666

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.512	0.102	161.900	44.473	0.000
poly(season_year_c, 3, raw = TRUE)1	-0.054	0.023	-2.381	1148.438	0.017
poly(season_year_c, 3, raw = TRUE)2	0.022	0.005	4.848	5237.468	0.000
poly(season_year_c, 3, raw = TRUE)3	-0.002	0.000	-6.689	5239.087	0.000
height_c	0.014	0.004	3.722	1669.492	0.000
positionDefender	-0.431	0.060	-7.231	1650.123	0.000
positionGoalkeeper	-1.240	0.098	-12.704	1666.856	0.000
positionMidfield	-0.228	0.060	-3.823	1677.849	0.000
citizen_of_club_countryYes	-0.399	0.038	-10.630	5729.204	0.000
RegionEast Asia & Pacific	-0.182	0.211	-0.861	1734.428	0.390
RegionLatin America & Caribbean	0.248	0.063	3.913	1796.512	0.000
RegionMiddle East & North Africa	0.083	0.245	0.340	1697.415	0.734
RegionNorth America	-0.005	0.225	-0.023	1778.504	0.981
RegionSub-Saharan Africa	0.018	0.104	0.176	1732.487	0.860
age_c	-0.001	0.004	-0.252	2354.791	0.801
goals	0.014	0.003	5.317	5538.686	0.000
assists	0.013	0.004	3.445	5075.156	0.001
red_cards	-0.019	0.030	-0.634	4696.827	0.526
yellow_cards	0.006	0.004	1.420	5406.162	0.156
minutes_played	0.511	0.017	29.411	5605.944	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.946
club_id	(Intercept)	0.369
club_id	season_year_c	0.033
Residual		0.614

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.636
club_id	21	0.096

Printing fit results for Model 3.2

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14507.097
BIC	14685.378

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.850	0.078	215.307	82.332	0.000
poly(season_year_c, 3, raw = TRUE)1	-0.056	0.017	-3.291	2124.850	0.001
poly(season_year_c, 3, raw = TRUE)2	0.028	0.003	7.947	5036.126	0.000
poly(season_year_c, 3, raw = TRUE)3	-0.002	0.000	-10.894	5037.446	0.000
height_c	0.010	0.004	2.695	1799.848	0.007
positionDefender	-0.404	0.058	-6.979	1786.194	0.000
positionGoalkeeper	-1.057	0.095	-11.159	1797.710	0.000
positionMidfield	-0.190	0.058	-3.301	1802.041	0.001
citizen_of_club_countryYes	-0.190	0.031	-6.039	6875.765	0.000
RegionEast Asia & Pacific	-0.431	0.204	-2.110	1836.234	0.035
RegionLatin America & Caribbean	0.191	0.061	3.133	1954.213	0.002
RegionMiddle East & North Africa	-0.092	0.237	-0.386	1814.616	0.699
RegionNorth America	-0.063	0.217	-0.292	1856.805	0.770
RegionSub-Saharan Africa	0.066	0.100	0.657	1848.911	0.511
poly(age_c, 2, raw = TRUE)1	0.019	0.004	4.988	2295.002	0.000
poly(age_c, 2, raw = TRUE)2	-0.024	0.000	-62.796	6152.216	0.000
goals	0.005	0.002	2.363	5169.192	0.018
assists	0.010	0.003	3.450	4882.603	0.001
red_cards	0.010	0.022	0.428	4678.452	0.669
yellow_cards	-0.002	0.003	-0.583	5089.437	0.560
minutes_played	0.353	0.013	26.260	5243.836	0.000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.960
club_id	(Intercept)	0.237
club_id	season_year_c	0.019
Residual		0.459

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.775
club_id	21	0.047

Printing fit results for Model 4

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14520.484
BIC	14712.479

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.857	0.079	212.455	77.432	0.000
poly(season_year_c, 3, raw = TRUE)1	-0.061	0.017	-3.550	1979.243	0.000
poly(season_year_c, 3, raw = TRUE)2	0.028	0.003	8.097	5037.191	0.000
poly(season_year_c, 3, raw = TRUE)3	-0.002	0.000	-10.969	5035.695	0.000
height_c	0.010	0.004	2.704	1797.210	0.007
positionDefender	-0.404	0.058	-6.991	1783.353	0.000
positionGoalkeeper	-1.057	0.095	-11.166	1795.098	0.000
positionMidfield	-0.190	0.058	-3.302	1799.253	0.001
citizen_of_club_countryYes	-0.186	0.031	-5.905	6884.155	0.000
RegionEast Asia & Pacific	-0.431	0.204	-2.113	1833.721	0.035
RegionLatin America & Caribbean	0.189	0.061	3.106	1951.968	0.002
RegionMiddle East & North Africa	-0.094	0.237	-0.395	1811.883	0.693
RegionNorth America	-0.066	0.217	-0.304	1854.349	0.761
RegionSub-Saharan Africa	0.064	0.100	0.637	1846.571	0.524
poly(age_c, 2, raw = TRUE)1	0.018	0.004	4.955	2294.634	0.000
poly(age_c, 2, raw = TRUE)2	-0.024	0.000	-62.790	6163.057	0.000
goals	0.005	0.002	2.386	5164.585	0.017
assists	0.010	0.003	3.473	4878.216	0.001
red_cards	0.009	0.022	0.391	4673.660	0.696
yellow_cards	-0.002	0.003	-0.572	5084.816	0.567
minutes_played	0.353	0.013	26.270	5239.334	0.000
uefa_coefficient	-0.001	0.001	-0.981	4466.845	0.327
foreign_players	0.199	0.097	2.053	1972.890	0.040

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.959
club_id	(Intercept)	0.244
club_id	season_year_c	0.019
Residual		0.459

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.773
club_id	21	0.050

LRT Statistic: 66.6861

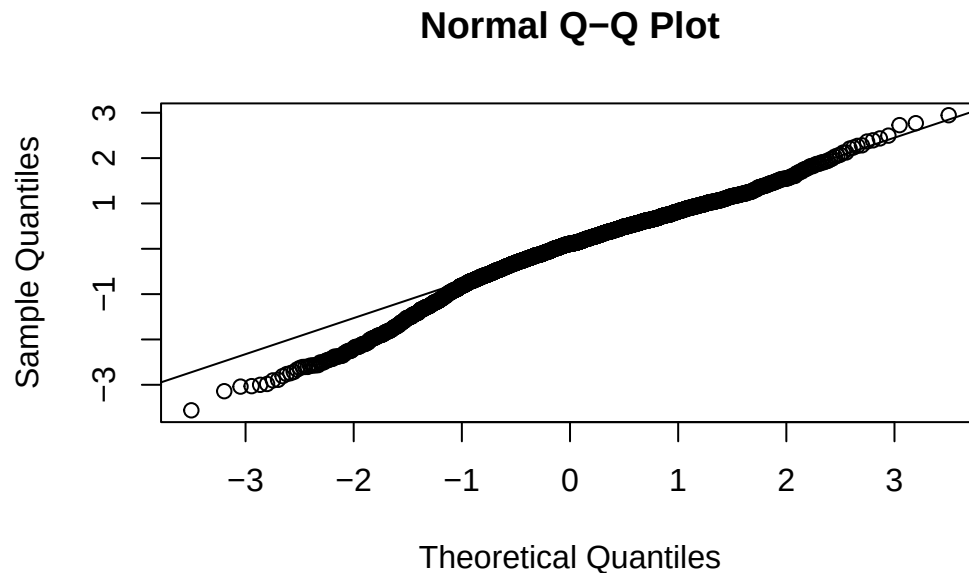
Parameter Diff: 2

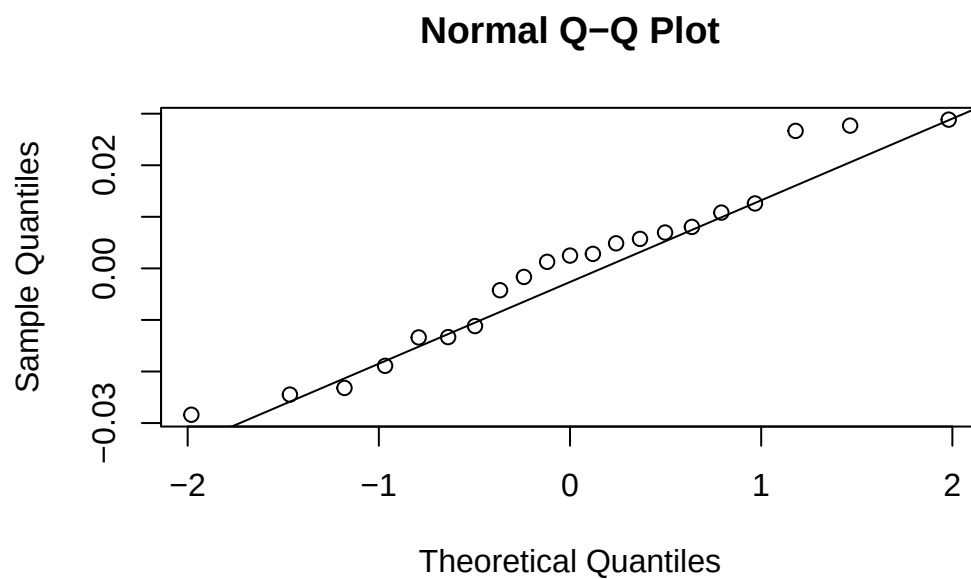
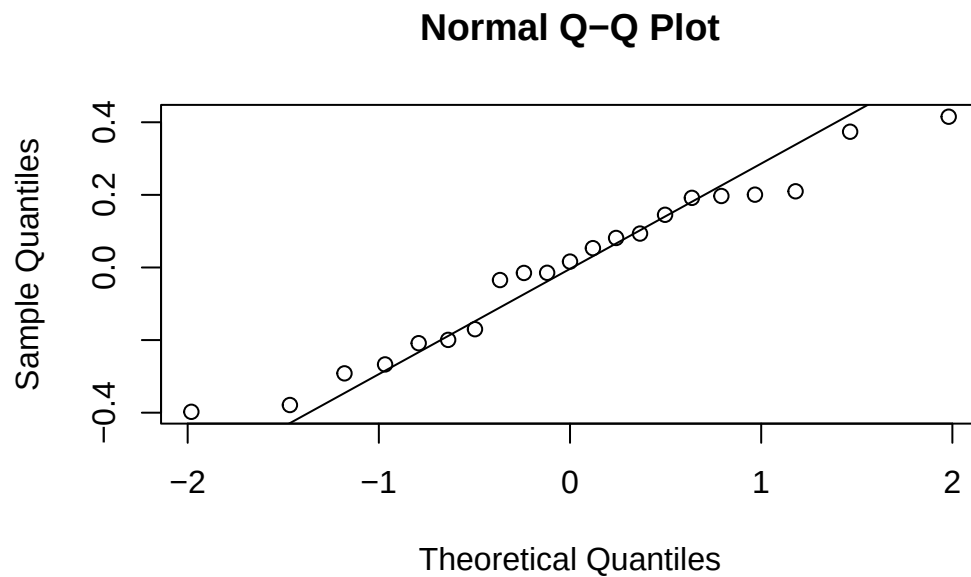
Distribution: Mixture of ChiSq(1) and ChiSq(2)

P-Value: < 0.0001

11 Appendix 4: Final Model Diagnostics

- By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.
- By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers. Thus, we cannot say the normal random effects assumption in the multilevel model are met.

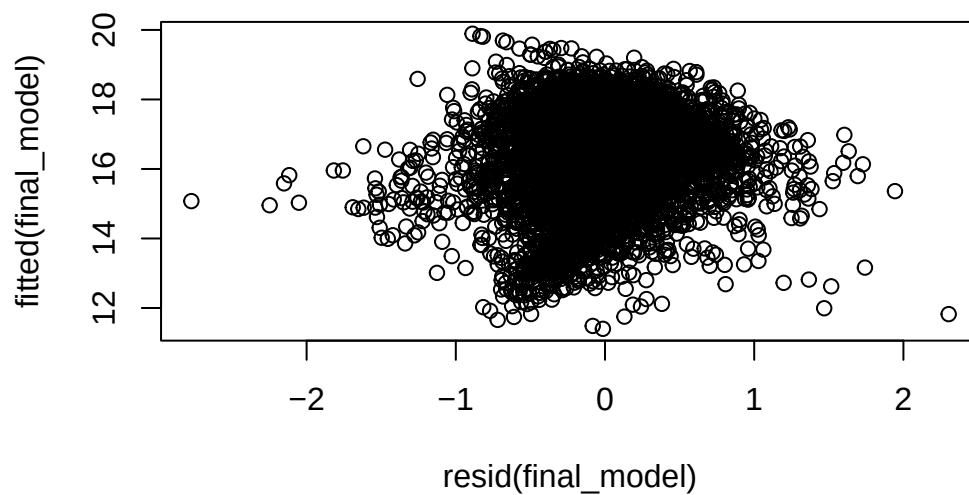
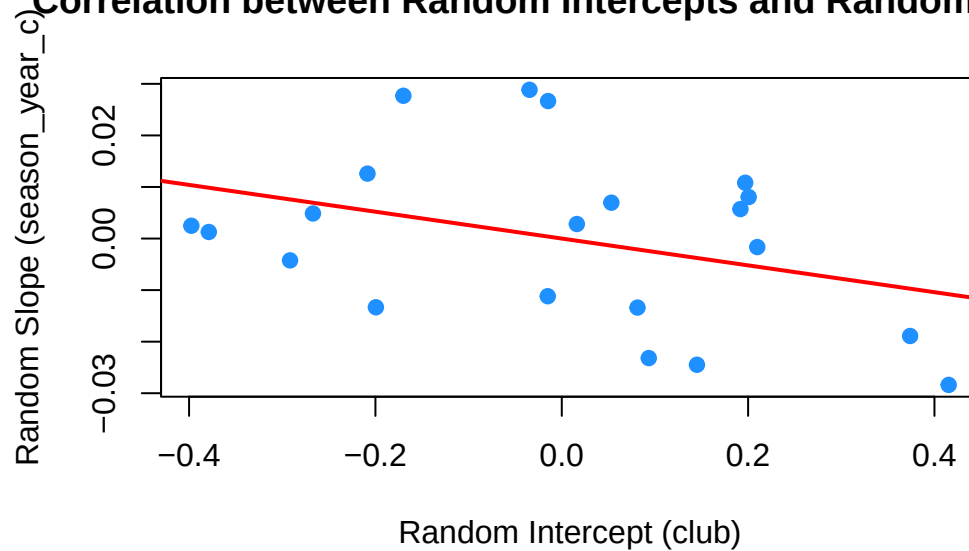




- The red regression line slopes slightly downward, but only gently. That is, clubs with higher baseline log market value tend to have slightly more negative or flatter slopes. Clubs with lower baseline market value tend to have slightly steeper slopes.
- Most residuals lie between about -1 and $+1$, and fitted log revenues between 14 and

18. There is no strong curved pattern or obvious systematic relationship between residual and fitted values. So the residual reasonably consistent with the linearity assumption. Besides, there is no funnel shape, so the residuals are homogeneous. However, we can still see some outliers exist.

Correlation between Random Intercepts and Random Slope



Model-predicted lines should resemble observed data lines



Reference line should be flat and horizontal



Reference line should be flat and horizontal



Points should be inside the contour lines



High collinearity (VIF) may inflate parameter uncertainty D



Dots should fall along the line



Dots should be plotted along the line



Dots should be plotted along the line



12 Appendix 5. AI Acknowledgement Statement

We relied on AI, specifically Claude, to assist us with the latex for our equation since at we had difficulty rendering the quarto file. Additionally, we sometimes used the assistance of AI to help us with our visualization, such as how removing specific grid lines in the background, or specific bold font. Finally, we had convergence issues specifically when attempting to introduce a random slope of year within players and we sought the wisdom of AI to come up with a solution below.